

Chemical properties of surface water (DP1.20093.001)

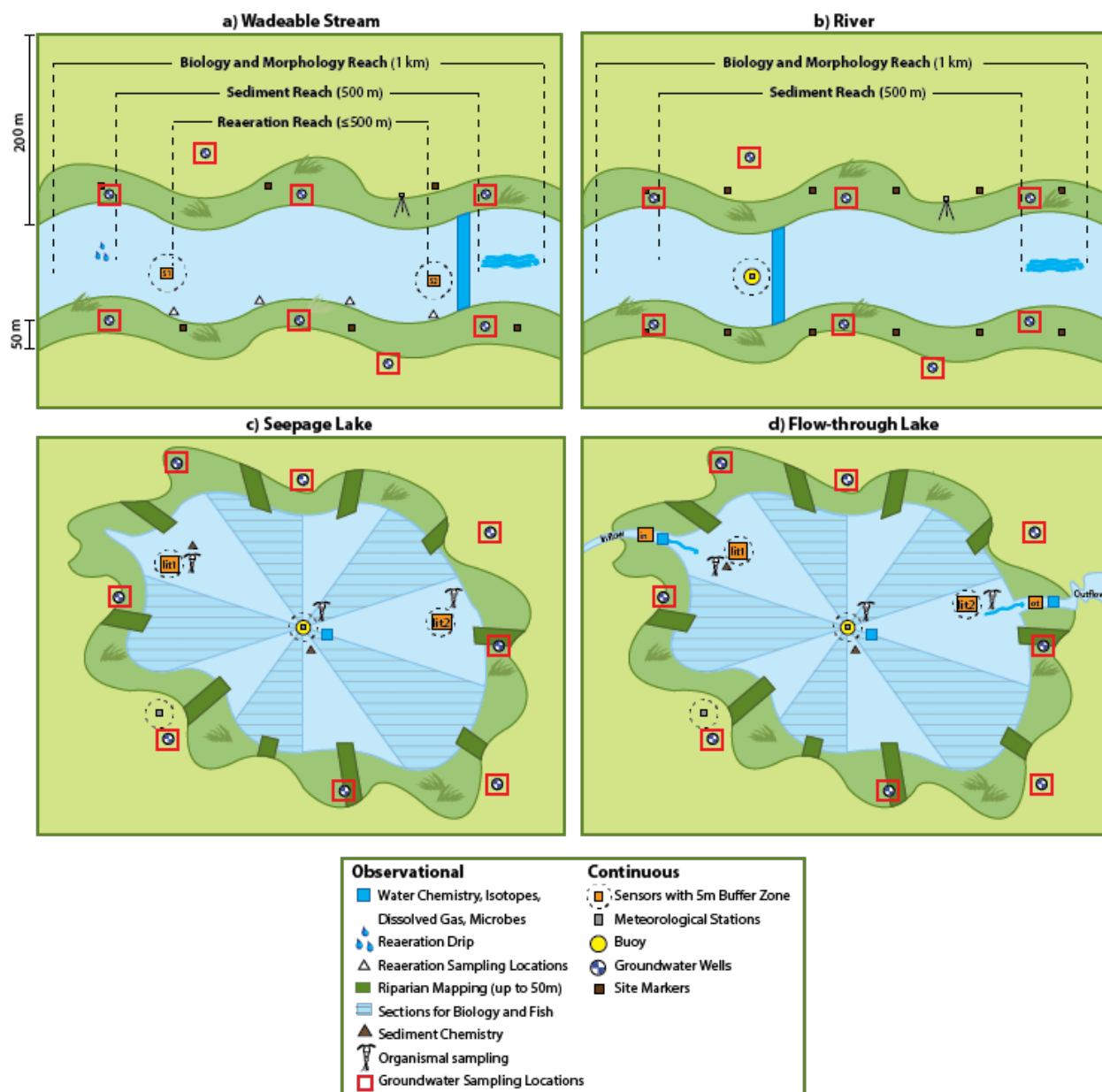
Measurement

Chemistry of surface water, including anions, cations, conductivity, pH, and various forms of carbon and nutrients including total, dissolved and particulates.

Collection methodology

Wadeable stream water collection occurs in the thalweg near the downstream sensor set while non-wadeable streams sample from a depth of 0.5m. Samples are taken up to 26 times per year (Fig 1). Lakes are sampled at up to 3 locations with variable depths near the bouy, inlet and outlet sensors 12 times per year (Fig 1). A more detailed description can be found in the AOS Protocol and Procedure: Water Chemistry Sampling in Surface Waters in Groundwater (NEON.DOC.002905). Samples are filtered within 3 hours of collection, kept cold, then shipped overnight to an analytical laboratory for processing.

For information about disturbances, land management activities, and other incidents that may impact data at NEON sites, see the [Site management and event reporting \(DP1.10111.001\)](#) data product.



Generic layout of NEON surface water sampling locations (blue quadrangles) in a) wadeable streams, b) rivers, c) seepage lakes, and d) flow-through lakes.

Data package contents

swc_fieldSuperParent: Field data for the parent sample of surface water chemistry

swc_asiPOMFieldData: Surface water stable isotopes field data

swc_externalLabSummaryData: Surface water chemistry external lab summary data

swc_externalLabAbsorbanceScan: Surface water chemistry absorbance scan from external lab

swc_domainLabData: Surface water chemistry ALK and ANC domain lab data summary data
swc_externalLabDataByAnalyte: Surface water chemistry from external lab in long format
swc_fieldData: Surface water chemistry summary data per site per bout
swc_externalLabBlanksByAnalyte: Blank sample water chemistry from external lab
variables: Description and units for each column of data in data tables
readme: Data product description, issue log, and other metadata about the data product
validation: Description of data validation applied at the points of collection and ingest

Data quality

Data on conditions during sampling can be found in the swc_fieldSuperParent table. The sampleCondition field in the swc_externalLabDataByAnalyte table provides information about the condition of the sample at the time of analysis. The swc_externalLabSummaryData table provides data about long-term performance of the analytical laboratory, including method detection limits, precision, and uncertainty.

Note that values can be blank (or NA when downloaded using neonUtilities) for analyteConcentration for a few different reasons. Users are encouraged to look at entries in sampleCondition and belowDetectionQF for details on the reason for a particular record. Records with entries other than "GOOD" or "OK" in sampleCondition likely indicate that data for a given analyte were not collected and users cannot infer a value for the analyte to include in their analysis. For records with "ND" values for belowDetectionQF, users can infer that the analyte concentration is very low and may want to include the records in their analysis.

A few parameters in this data product are also available via the in-stream sensors, most notably dissolved oxygen, specific conductance, and pH. Given the potential for additional uncertainty arising from sample collection/processing, transport, and storage time prior to analysis, we strongly recommend using the in-situ Water quality (DP1.20288.001) sensor data whenever possible, particularly for pH. Alternatively, alkalinity and acid neutralizing capacity (ANC) titrations are typically performed within 24 hours of sample collection and the initialSamplepH found in the swc_domainLabData expanded package is another option for users when sensor data is not available. When comparing water chemistry grab samples to sensor-based water quality data (DP1.20288.001), note that water chemistry samples are collected at the downstream sensor set (horizontalPosition 102 or 112) in streams and at the buoy location (horizontalPosition 103) in rivers and lakes. Because of lower uncertainty, we encourage the use of continuous sensor data whenever possible.

Please note that quality checks are comprehensive but not exhaustive; therefore, unknown data quality issues may exist. Users are advised to evaluate quality of the data as relevant to the scientific research question being addressed, perform data review and post-processing prior to analysis, and use the data quality information and issue logs included in download packages to aid interpretation.

Standard calculations

For wrapper functions to download data from the API, and functions to merge tabular data files across sites and months, NEON provides the neonUtilities package in R and the neonutilities package in Python. See the [Download and Explore NEON Data](#) tutorial for introductory instructions in both programming languages.

Note that data in the swc_externalLabDataByAnalyte table are reported in long format, with one record for each combination of sample and analyte.

Table joining

Table 1	Table 2	Join by field(s)
swc_fieldSuperParent	swc_fieldData	parentSampleID
swc_fieldData	swc_domainLabData	parentSampleID
swc_fieldData	swc_externalLabDataByAnalyte	Not fully automatable: sampleID in swc_externalLabDataByAnalyte corresponds to filtSampleID, filtNutSampleID, rawSampleID, rawNutSampleID, and pcnSampleID in swc_fieldData
swc_asipomFieldData	swc_domainLabData	parentSampleID
swc_asipomFieldData	swc_externalLabDataByAnalyte	Not fully automatable: sampleID in swc_externalLabDataByAnalyte corresponds to isotopePOMSampleID or isotopePOMRep2SampleID in swc_asipomFieldData
swc_asipomFieldData	swc_fieldData	parentSampleID
swc_asipomFieldData	swc_fieldSuperParent	parentSampleID
swc_domainLabData	swc_externalLabDataByAnalyte	Requires intermediate table: join via swc_fieldData table
swc_domainLabData	swc_fieldSuperParent	parentSampleID
swc_externalLabDataByAnalyte	swc_fieldSuperParent	Requires intermediate table: join via swc_fieldData table

Table 1	Table 2	Join by field(s)
swc_externalLabSummaryData	Any other table	Join not recommended. Relevant quality control data can be connected to analytes by overlap of analysisDate with labSpecific start and end dates.

Documentation



[NEON Aquatic Sampling Strategy](#)

NEON.DOC.001152vD | 846.5 KiB | PDF



[AOS Protocol and Procedure: SWC – Water Chemistry Sampling in Surface Waters and Groundwater](#)

NEON.DOC.002905vK | 3.8 MiB | PDF



[NEON Algorithm Theoretical Basis Document \(ATBD\): OS Data Quality Control](#)

NEON.DOC.005424vA | 651.3 KiB | PDF



[NEON User Guide to Chemical Properties of Surface Water \(NEON.DP1.20093\) and Chemical Properties of Groundwater \(NEON.DP1.20092\)](#)

NEON_waterChem_userGuide_vH | 832.1 KiB | PDF

For more information on data product documentation, see:
<https://data.neonscience.org/data-products/DP1.20093.001>

Citation

To cite data from value: "Chemical properties of surface water" (DP1.20093.001), see citation here:
<https://data.neonscience.org/data-products/DP1.20093.001>

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<https://www.neonscience.org/data-samples/guidelines-policies/citing>

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